Heart Disease Risk Prediction Using Logistic Regression and Support Vector Machines

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***Abstract*—Heart disease is one of the main causes of death around the world, so predicting it early is very important. In this study, we use machine learning to predict whether a patient has heart disease based on basic medical information such as age, sex, chest pain type, blood pressure, cholesterol, and ECG results. We treat this as a binary classification problem (having heart disease or not) and use the Cleveland Heart Disease dataset. A simple machine learning pipeline is built, including data preprocessing, model training, and evaluation. We test several models, including Logistic Regression (L1 and L2) and Support Vector Classifier (linear and RBF). The models are evaluated using common metrics such as Accuracy, Precision, Recall, F1-score, and ROC-AUC to compare their performance. In addition, we further analyzed the models by adjusting the prediction threshold to achieve a balance between prediction accuracy and cost of additional tests. We especially focused on reducing the number of missed disease cases. We also checked whether the predicted probabilities from the models are reliable by comparing them with actual outcomes. Finally, we interpreted the Logistic Regression model to understand which factors increase or decrease the risk of heart disease. The results show that features such as chest pain type, the number of blocked blood vessels, and some specific test results have a strong impact on the likelihood of having the disease.**

***Index Terms*—heart disease prediction, machine learning, logistic regression, support vector machine, binary classification, threshold tuning, model calibration, healthcare analytics, Cleveland Heart Disease dataset**

I. Introduction

Cardiovascular disease is one of the leading causes of death worldwide and remains a major challenge for modern healthcare systems. Early detection of individuals at risk of heart disease is important for improving patient outcomes and enabling timely medical treatment. With the rapid growth of healthcare data, machine learning has become a useful tool for supporting disease prediction and assisting clinical decision-making.

Heart disease prediction can be treated as a binary classification problem based on patient clinical information such as age, blood pressure, cholesterol level, and electrocardiographic results. However, medical datasets often contain noise, missing values, and complex relationships between features, which make the prediction task more challenging. Many previous studies have applied different machine learning algorithms for heart disease prediction. Among them, Logistic Regression and Support Vector Machine are commonly used due to their effectiveness in binary classification tasks and their ability to work well with structured tabular data. However, most studies mainly focus on model accuracy, while less attention is given to probability reliability and decision threshold selection, which are important in real-world medical applications.

In this study, we use the Cleveland Heart Disease dataset to build a machine learning model for predicting heart disease risk. A complete pipeline is developed, including data preprocessing, feature transformation, model training, and performance evaluation. In addition to comparing model performance using common metrics such as Accuracy, Precision, Recall, F1-score, ROC-AUC, and PR-AUC, we further analyze the models by adjusting the prediction threshold to balance prediction performance and practical cost. We also evaluate the reliability of predicted probabilities to ensure that the model outputs can be trusted in real-world scenarios. The objective of this study is to provide not only a performance comparison of machine learning models but also practical insights into how these models can be applied effectively in healthcare screening systems.

II. Related Work

A. Logistic Regression

Logistic Regression is a supervised machine learning algorithm used for binary classification problems. It models the probability that an input sample belongs to a particular class (usually class 1) based on the input features.

First, the model computes a linear combination of the input features:

$$z = w^T x + b \tag{1}$$

Then, the sigmoid function is applied:

$$\sigma(z) = \frac{1}{1 + e^{-z}} \tag{2}$$

where $\sigma(z)$ transforms the linear output $z \in (-\infty, +\infty)$ into a probability value in the range $(0, 1)$, making it suitable for binary classification.

The final prediction is obtained using a threshold τ (commonly 0.5):

$$\hat{y} = \begin{cases} 1, & \text{if } \sigma(z) \geq \tau \\ 0, & \text{otherwise} \end{cases} \tag{3}$$

During training, Logistic Regression minimizes the log-loss function:

$$\mathcal{L} = -\frac{1}{N}\sum_{i=1}^{N}\left[y_i \log(\hat{y}_i) + (1-y_i)\log(1-\hat{y}_i)\right] \tag{4}$$

The model parameters are optimized using Gradient Descent:

$$w = w - \alpha\frac{\partial \mathcal{L}}{\partial w} \tag{5}$$

$$b = b - \alpha\frac{\partial \mathcal{L}}{\partial b} \tag{6}$$

where α is the learning rate. The parameters w and b are updated iteratively until the model converges to an optimal solution.

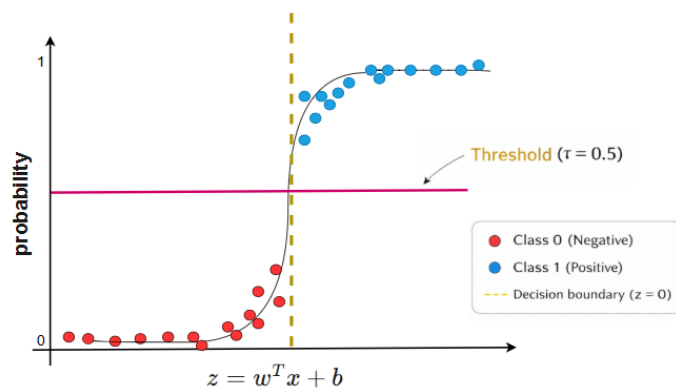

Fig. 1: Logistic Regression Sigmoid Function and Classification Threshold

Logistic Regression is often used as a simple baseline model for binary classification because it is fast and easy to train. Instead of just giving a final prediction (0 or 1), it also outputs the probability that a sample belongs to class 1, which allows us to choose different decision thresholds depending on the task. Another advantage is that Logistic Regression is easy to interpret. Each feature has a coefficient, so we can see how much each factor affects the prediction. However, this model works best when the data can be separated by a straight line (a linear boundary). If the relationship between the features and the target is more complex (nonlinear), Logistic Regression may not perform well because it cannot capture those patterns. In addition, it does not automatically learn interactions between features, so we often need to create additional features manually to improve performance.

B. SVC (Support Vector Classifier)

Support Vector Classifier (SVC) is a supervised machine learning algorithm commonly used for binary classification tasks. The main idea of SVC is to find an optimal separating hyperplane that divides data points belonging to different classes in the feature space. The algorithm aims to maximize the margin, which is the distance between the decision boundary and the closest data points from each class. These closest data points are called support vectors, and they play an important role in defining the position of the decision boundary. A larger margin generally leads to better generalization performance when predicting unseen data. Similar to Logistic Regression, Linear SVC produces a decision score for each input sample. The model computes a linear function:

$$z = w^T x + b \tag{7}$$

where x represents the input feature vector, w is the weight vector, and b is the bias term. Although this equation has the same form as Logistic Regression, the interpretation is different. In SVC, the value z represents the signed distance (decision score) from the sample to the separating hyperplane rather than a probability value. Therefore, it cannot be interpreted as a probability. Intuitively, a larger absolute value $|z|$ means that the sample is farther away from the decision boundary and the model is more confident about the prediction. The predicted class is determined based on the sign of z:

$$\hat{y} = \begin{cases} 1, & z \geq 0 \\ 0, & z < 0 \end{cases} \tag{8}$$

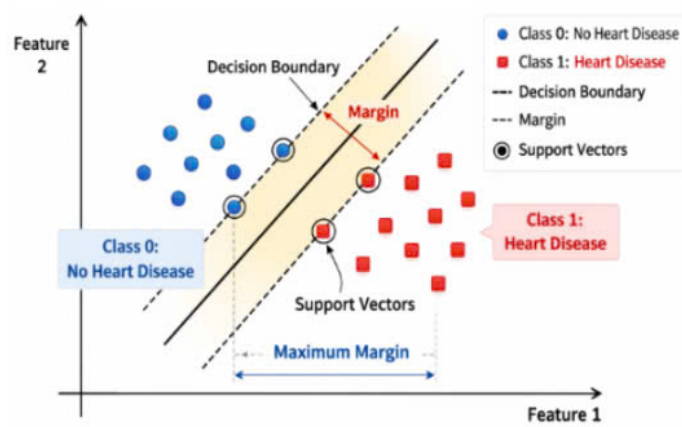

Fig. 2: Support Vector Machine (SVM) Decision Boundary and Margin

SVC is a margin-based classifier that often performs well on structured tabular datasets such as medical data. By maximizing the separation margin, the model can achieve strong generalization performance in binary classification tasks. In addition, kernelized SVC (e.g., using the RBF kernel) can capture non-linear relationships between features and class labels, which is useful when the relationship between clinical features and disease outcomes is complex. However, the raw output of SVC is a decision score rather than a calibrated probability, so it cannot be directly interpreted as a risk probability without an additional calibration step. Furthermore, SVC, especially with non-linear kernels, can be less interpretable than linear models such as Logistic Regression, and its performance may be sensitive to hyperparameters.

III. Main Contributions

This study addresses three main research questions. First, we compare Logistic Regression and Support Vector Classifier (SVC) to determine which model provides better prediction performance based on evaluation metrics. Second, we examine how different decision thresholds, particularly those based on recall and precision, affect classification outcomes. Finally, we analyze the calibration of predicted probabilities to assess whether these values can be reliably interpreted as risk scores for heart disease. To answer these questions, a complete machine learning pipeline is developed, including data preprocessing, model training, and evaluation. The dataset is divided into training and testing sets, and models such as Logistic Regression and SVC are trained on the training data. Model performance is evaluated using metrics such as ROC-AUC, PR-AUC, and F1-score. In addition, different classification thresholds and probability calibration techniques are explored to better understand model behavior.

A. Problem Formulation

In this study, the heart disease prediction task is formulated as a binary classification problem using the Cleveland Heart Disease dataset. The dataset contains 303 patient records with 13 clinical features, including age, sex, chest pain type, resting blood pressure, cholesterol level, electrocardiographic results, maximum heart rate, and other medical indicators. In the original dataset, the target variable is defined with five classes (0–4) representing different levels of heart disease severity. In this study, the problem is simplified into a binary classification task defined as:

$$y = \begin{cases} 0, & \text{no heart disease} \\ 1, & \text{presence of heart disease} \end{cases} \tag{9}$$

The objective of the model is to learn a mapping function:

$$f(x) = P(y = 1 \mid x) \tag{10}$$

where x is the patient's feature vector and $P(y = 1 \mid x)$ is the predicted probability of heart disease. The final prediction is made using a classification threshold.

$$\hat{y} = \begin{cases} 1, & P(y = 1 \mid x) \geq \tau \\ 0, & \text{otherwise} \end{cases} \tag{11}$$

The threshold τ is important because it controls how the model makes decisions. If we choose a lower threshold, the model will detect more patients with heart disease, but it may also wrongly label some healthy people as sick. On the other hand, if we choose a higher threshold, the model will make fewer false alarms, but it may miss some real disease cases. In this study, we do not only use the default threshold ($\tau = 0.5$). Instead, we test different threshold values to find a better balance between detecting disease and avoiding unnecessary follow-up tests. This is especially important in healthcare, where missing a disease case can be very serious. This also shows why it is important for the model's predicted probabilities to be reliable. If the probabilities are accurate, then choosing a threshold will lead to better and more trustworthy decisions.

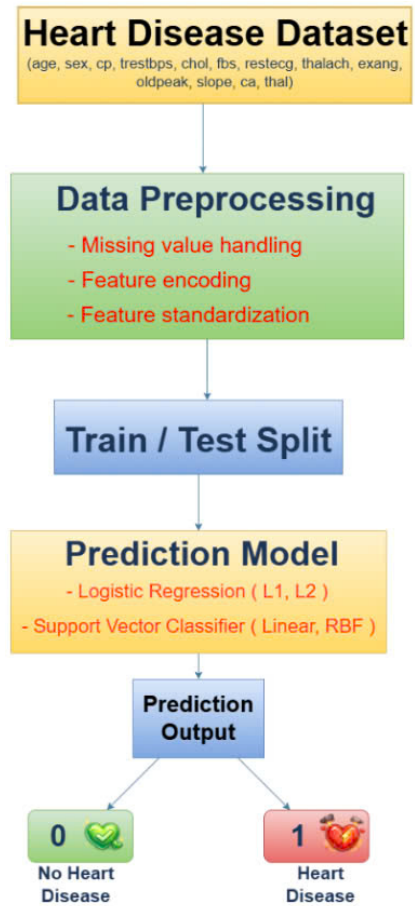

Fig. 3: Overview of the heart disease prediction framework

B. Data Engineering

In this study, data preprocessing is performed before training the machine learning models to ensure data quality and improve model performance. First, the dataset is divided into two subsets: 80% for training and 20% for testing. A stratified split is applied to preserve the distribution of the target variable in both sets. The training set is used to fit the models, while the test set is used to evaluate their performance on unseen data. After splitting the dataset, several preprocessing steps are applied. For numerical features, missing values are handled using median imputation to reduce the effect of outliers. Then, the features are standardized using StandardScaler so that all variables have a similar scale. This step is important because many machine learning algorithms are sensitive to the scale of input features. For categorical features, missing values are replaced using the most frequent value. After that, One-Hot Encoding is applied to convert categorical variables into numerical form. This allows machine learning models to process categorical information effectively without introducing

ordinal relationships between categories. All preprocessing steps are combined into a unified pipeline to ensure consistent transformation of both training and testing data. This approach helps prevent data leakage and improves the reliability of model evaluation.

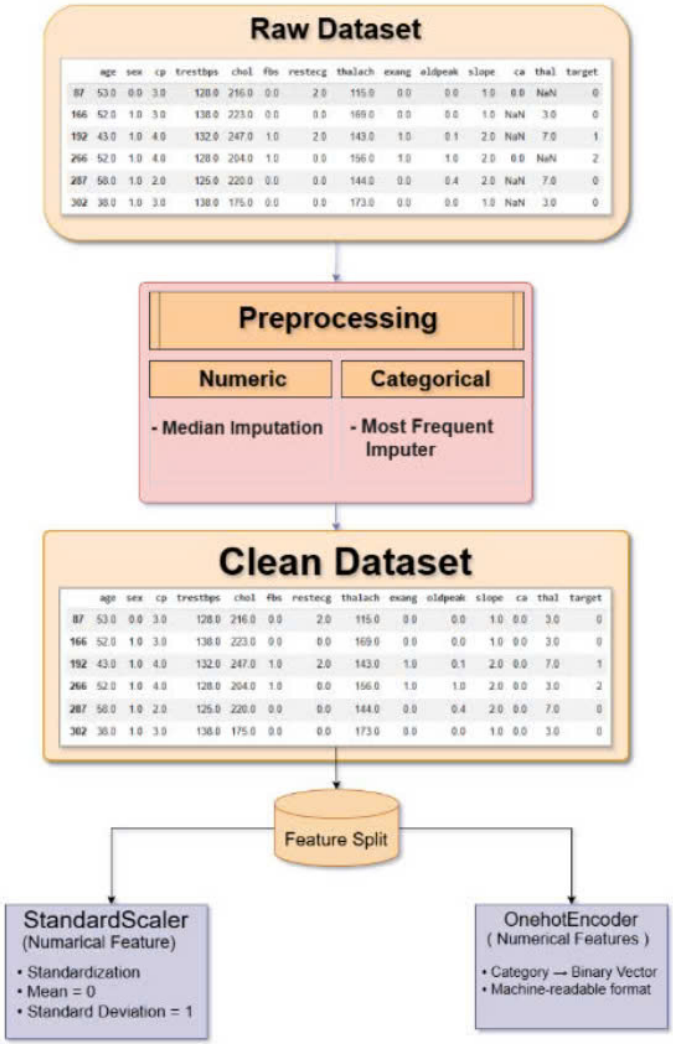

Fig. 4: Data preprocessing pipeline used in the proposed framework

C. Model Design

In this study, several classification models are employed to predict the presence of heart disease based on patient clinical attributes. The selected models include a baseline model and several widely used machine learning algorithms in order to evaluate their effectiveness in the heart disease prediction task. Specifically, the models used in this study include the Majority classifier, Logistic Regression with L1 and L2 regularization, and Support Vector Classifier (SVC) with linear and RBF kernels. All models are trained and tested in the same way so the comparison is fair. We also look at different evaluation metrics and try different threshold values to better understand how the models behave in real-world situations.

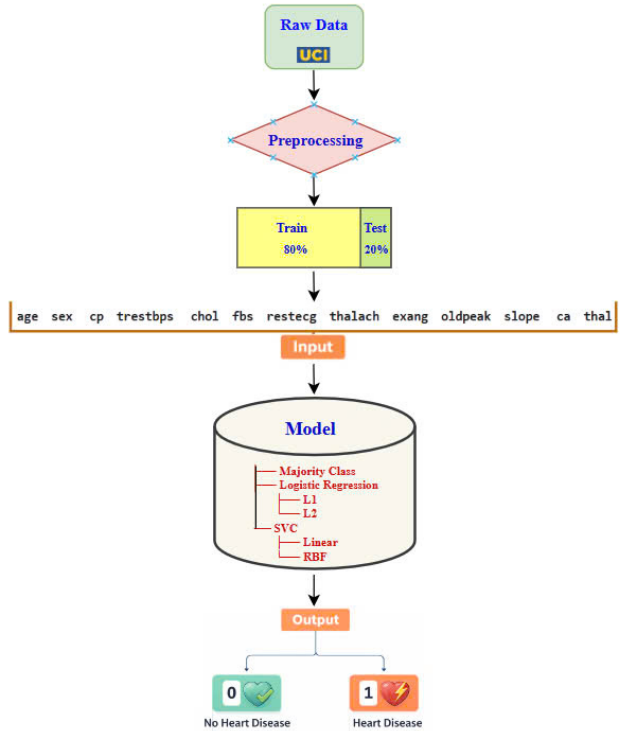

Fig. 5: Overall architecture of the classification models.

*1) **Majority Baseline**:* The Majority classifier is used as a baseline model. It simply predicts the most frequent class in the training dataset for all samples. This baseline helps determine whether more advanced models can achieve better performance. Fig. 6 shows an example where class 0 is the most frequent label in the training dataset, leading the Majority Baseline to always predict class 0 for any input.

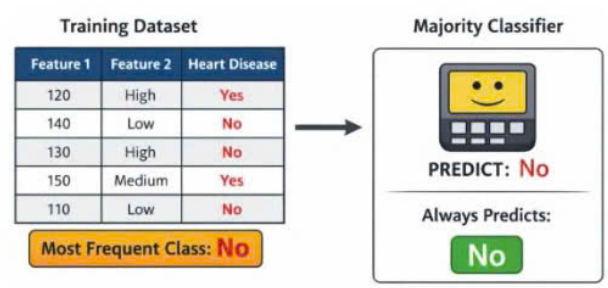

Fig. 6: Majority Classifier Predicting the Most Frequent Class

*2) **Logistic Regression**:* Logistic Regression is a widely used linear classification algorithm for binary classification problems. The model estimates the probability that a sample belongs to the positive class using the logistic (sigmoid) function.

The logistic function is defined as:

$$P(y = 1 \mid x) = \sigma(z) = \frac{1}{1 + e^{-z}} \tag{12}$$

$$z = w^T x + b \tag{13}$$

where, x represents the feature vector of a patient, w is the weight vector, and b is the bias term. The output of the model is a probability value in the range $(0, 1)$ that represents the likelihood of heart disease. The final prediction is obtained by applying a classification threshold:

$$\hat{y} = \begin{cases} 1, & \text{if } P(y = 1 \mid x) \geq \tau \\ 0, & \text{otherwise} \end{cases} \tag{14}$$

In this study, two variants of Logistic Regression are considered to examine the effect of regularization on model performance.

Logistic Regression with L1 Regularization

L1 regularization adds a penalty term to the loss function that encourages sparsity in the model weights. The objective function can be written as:

$$\mathcal{L}(w, b) = -\frac{1}{N} \sum_{i=1}^{N} [y_i \log(\hat{y}_i) + (1 - y_i) \log(1 - \hat{y}_i)] + \lambda \|w\|_1 \tag{15}$$

where $\|w\|_1 = \sum |w_j|$ is the L1 norm and λ controls the strength of regularization. L1 regularization encourages sparse weights, which can effectively perform feature selection and reduce model complexity.

Logistic Regression with L2 Regularization

L2 regularization introduces a penalty term that discourages large weight values. The objective function is defined as:

$$\mathcal{L}(w, b) = -\frac{1}{N} \sum_{i=1}^{N} [y_i \log(\hat{y}_i) + (1 - y_i) \log(1 - \hat{y}_i)] + \lambda \|w\|_2^2 \tag{16}$$

where $\|w\|_2^2 = \sum w_j^2$ is the squared L2 norm.

Unlike L1 regularization, L2 regularization does not force weights to become exactly zero but instead shrinks them toward smaller values. This helps prevent overfitting and improves generalization performance.

*3) **Support Vector Classifier (SVC)**:* Support Vector Classifier (SVC) is a supervised machine learning algorithm commonly used for binary classification tasks. The main idea of SVC is to find an optimal separating hyperplane that divides data points belonging to different classes in the feature space. The algorithm aims to maximize the margin, which is the distance between the decision boundary and the closest data points from each class. These closest data points are called support vectors and play a crucial role in defining the decision boundary.

SVC with Linear Kernel

For the linear kernel, the decision function of SVC is defined:

$$f(x) = w^T x + b \tag{17}$$

where x is the input feature vector, w is the weight vector, and b is the bias term. The predicted class label is determined based on the sign of the decision function:

$$\hat{y} = \begin{cases} 1, & \text{if } f(x) \geq 0 \\ 0, & \text{otherwise} \end{cases} \tag{18}$$

The objective of SVC is to maximize the margin while minimizing classification errors. This can be formulated as the following optimization problem:

$$\min_{w,b} \frac{1}{2} \|w\|^2 \tag{19}$$

subject to:

$$y_i(w^T x_i + b) \geq 1, \quad \forall i \tag{20}$$

SVC with RBF Kernel

To capture nonlinear relationships, SVC can use kernel functions to implicitly map data into a higher-dimensional feature space. In this study, the Radial Basis Function (RBF) kernel is used.

The RBF kernel is defined as:

$$K(x_i, x_j) = \exp(-\gamma \|x_i - x_j\|^2) \tag{21}$$

where γ is a hyperparameter that controls the influence of each training sample.

By using the kernel trick, SVC can construct nonlinear decision boundaries without explicitly transforming the data into a higher-dimensional space. This allows the model to effectively capture complex patterns in the dataset.

In this study, both linear and RBF kernel SVC models are trained after applying preprocessing steps such as missing value handling, feature scaling, and categorical encoding.

D. Evaluation Framework

To evaluate the performance of the classification models in the heart disease prediction task, several commonly used evaluation metrics for binary classification problems are employed. These metrics provide a comprehensive assessment of the predictive performance of each model on the test dataset.

First, the confusion matrix is used to analyze the prediction results of the models. The confusion matrix consists of four main components:

- True Positive (TP): the number of patients with heart disease that are correctly predicted as positive.
- True Negative (TN): the number of patients without heart disease that are correctly predicted as negative.

- False Positive (FP): the number of patients without the disease that are incorrectly predicted as having the disease.
- False Negative (FN): the number of patients with the disease that are incorrectly predicted as negative.

Based on these values, several important evaluation metrics are calculated to assess model performance.

Accuracy

Accuracy measures the proportion of correctly predicted samples over the total number of samples in the test dataset. It is defined as:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \tag{22}$$

This metric provides an overall measure of prediction correctness.

Precision

Precision measures the proportion of correctly predicted positive samples among all predicted positive samples.

$$\text{Precision} = \frac{TP}{TP + FP} \tag{23}$$

This metric indicates how many of the predicted heart disease cases are actually correct.

Recall (also known as Sensitivity) measures the ability of the model to correctly identify patients with heart disease.

$$\text{Recall} = \frac{TP}{TP + FN} \tag{24}$$

In medical prediction tasks, recall is particularly important because missing a disease case may lead to serious consequences.

F1-score

The F1-score is the harmonic mean of precision and recall:

$$\text{F1-score} = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} \tag{25}$$

This metric provides a balanced evaluation when both precision and recall are important.

ROC Curve and AUC

In addition to the above metrics, this study uses the Receiver Operating Characteristic (ROC) curve to evaluate the classification performance of the models across different thresholds. The ROC curve represents the relationship between the True Positive Rate (TPR) and the False Positive Rate (FPR).

$$\text{TPR} = \frac{TP}{TP + FN} \tag{26}$$

$$\text{FPR} = \frac{FP}{FP + TN} \tag{27}$$

E. Training and Testing Strategy

To ensure a fair and reliable evaluation of the proposed models, the dataset is divided into training and testing sets. After converting the original target variable into a binary classification problem, the dataset is split using an 80:20 ratio, where 80% of the data is used for training and 20% is reserved for testing. A stratified split is applied to preserve the class distribution across both subsets.

During the training phase, the data is processed through a unified preprocessing pipeline to ensure consistent data transformation. Numerical features are handled using median imputation followed by feature standardization using StandardScaler, while categorical features are processed using most-frequent imputation and OneHotEncoder. After preprocessing, multiple classification models are trained on the training set, including the Majority classifier, Logistic Regression with L1 and L2 regularization, and Support Vector Classifier (SVC) with linear and RBF kernels.

To obtain robust performance estimates and reduce the risk of overfitting, Stratified 5-fold cross-validation is applied during the training stage. After the training process, models are evaluated on the independent test set. The predictive performance of the models is measured using several evaluation metrics, including Accuracy, Precision, Recall, F1-score, ROC-AUC, and Average Precision (AP). Additionally, ROC curves and Precision–Recall curves are used to visualize the classification performance across different decision thresholds.

IV. Experimental Results

A. Dataset

Dataset Description

The experiments in this study are conducted using the Cleveland Heart Disease dataset, which is a widely used benchmark dataset for cardiovascular disease prediction. The dataset contains 303 patient samples with 13 clinical attributes describing various physiological measurements and medical indicators.

These attributes include demographic and medical information such as age, sex, chest pain type, resting blood pressure, cholesterol level, fasting blood sugar, resting electrocardiographic results, maximum heart rate achieved, exercise-induced angina, ST depression, slope of the ST segment, number of major vessels colored by fluoroscopy, and thalassemia.

In the original dataset, the target variable contains five classes (0–4) representing the severity of heart disease. In this study, the problem is reformulated as a binary classification task, defined as follows:

- 0: indicates the absence of heart disease
- 1: indicates the presence of heart disease

This transformation simplifies the prediction task and focuses on detecting whether a patient is likely to have heart disease.

Dataset Visualization

To better understand the dataset characteristics, exploratory data analysis (EDA) is performed using multiple visualization techniques.

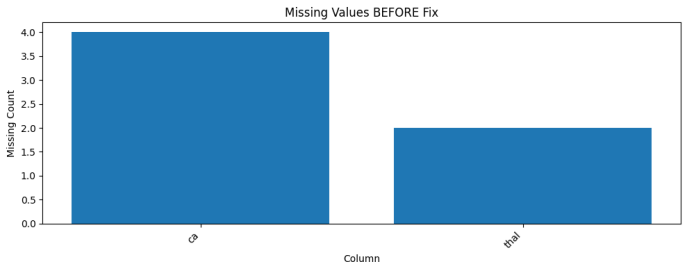

Fig. 7: Missing value distribution before preprocessing

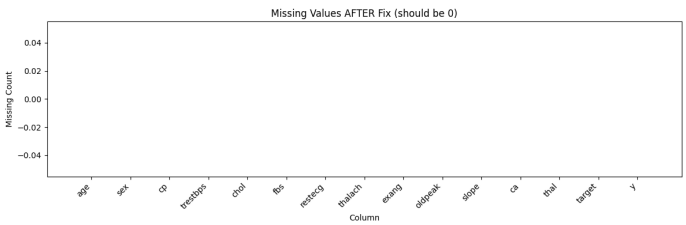

Fig. 8: Missing value distribution after preprocessing

These figures illustrate the distribution of missing values in the dataset before and after preprocessing. Missing values are primarily observed in the ca and $thal$ attributes.

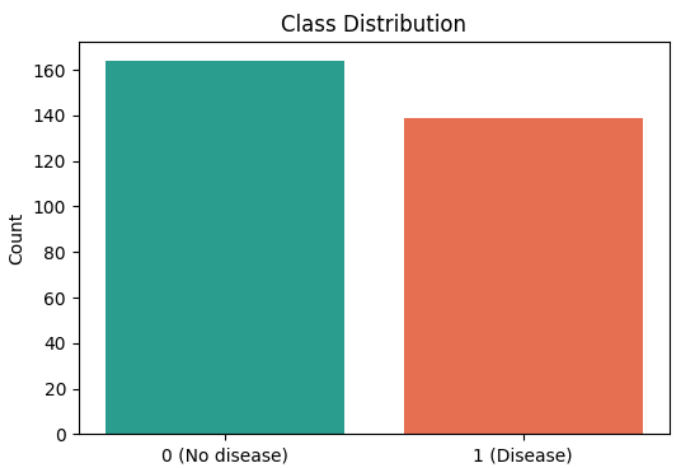

Fig. 9: Distribution of the target variable

The figure presents the distribution of patients with and without heart disease. The dataset shows relatively balanced classes, which is beneficial for training classification models. This helps reduce bias toward one class and allows the model to learn patterns from both groups more effectively. As a result, the evaluation metrics can better reflect the true performance of the models. This balanced distribution is especially important in medical applications, where both classes need to be predicted accurately.

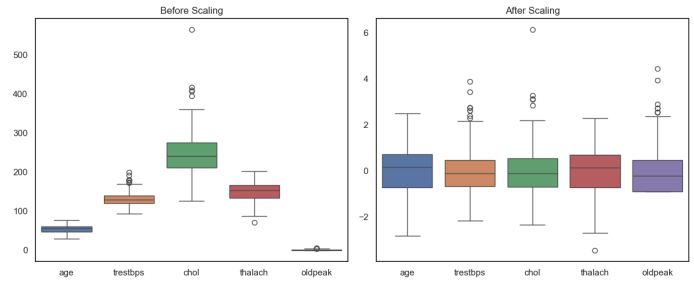

Fig. 10: Boxplot visualization of numerical attributes

Although several outliers (Fig.10) are observed in variables such as blood pressure, cholesterol, and ST depression, we decided to retain the outliers because they represent serious pathological cases that cannot be ignored in cardiovascular diagnosis. To prevent these extreme values from skewing the model, we used StandardScaler and OneHotEncoder scale the data to the same range. This approach allows the model to both utilize important pathological markers and maintain high accuracy during training.

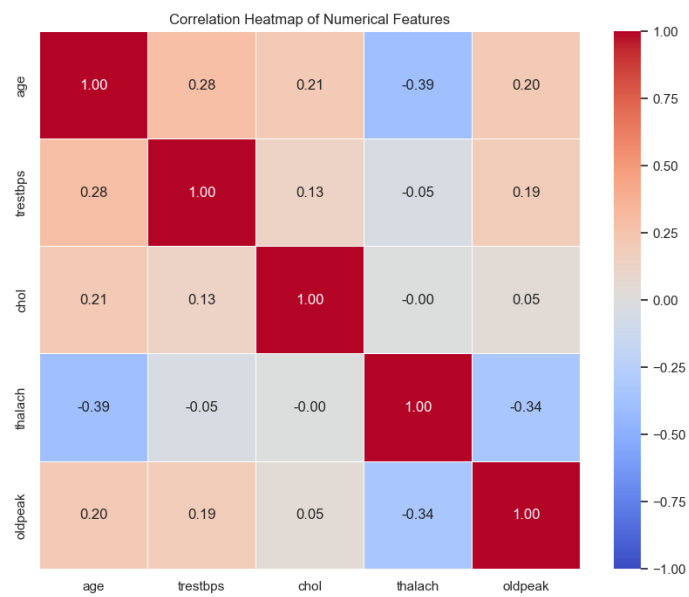

Fig. 11: Correlation heatmap of numerical features

From the heatmap, we can see a clear negative relationship between age and maximum heart rate (thalach) (-0.39), meaning heart performance tends to decrease as people get older. Blood pressure and cholesterol also increase slightly with age, but the relationship is not very strong.

Overall, the variables are not too closely related to each other, so they don't repeat the same information much (no multicollinearity). This allows machine learning models to learn information from each variable more efficiently.

B. Result Comparison

To evaluate the proposed approach, several classification models are trained and compared, including the Majority classifier, Logistic Regression (L1, L2), and Support Vector Classifier (linear, RBF). Model performance is assessed using Accuracy, Precision, Recall, F1-score, and ROC-AUC.

Model	Accuracy	Precision	Recall	F1-score	AUC
LogReg_L2	**0.885246**	**0.838710**	**0.928571**	**0.881356**	**0.966450**
SVC_RBF	**0.885246**	**0.838710**	**0.928571**	**0.881356**	0.964286
SVC_Linear	0.852459	0.787879	**0.928571**	0.852459	0.961039
LogReg_L1	0.868852	0.833333	0.892857	0.862069	0.958874
Majority	0.540984	0.000000	0.000000	0.000000	0.500000

TABLE I: Model Performance Comparison

As shown in Table 1, the Majority classifier performs poorly, with Accuracy (0.541), Precision (0.000), Recall (0.000), and F1-score (0.000), showing that it cannot identify heart disease cases.

Logistic Regression (L2) achieves the best performance, with high Accuracy (0.885), Recall (0.929), and F1-score (0.881), along with the highest AUC (0.966), indicating strong and reliable predictions.

Logistic Regression (L1) and SVC perform slightly worse than L2. Overall, the models show a good balance between Accuracy and Recall, making them suitable for heart disease prediction.

C. Result Analysis

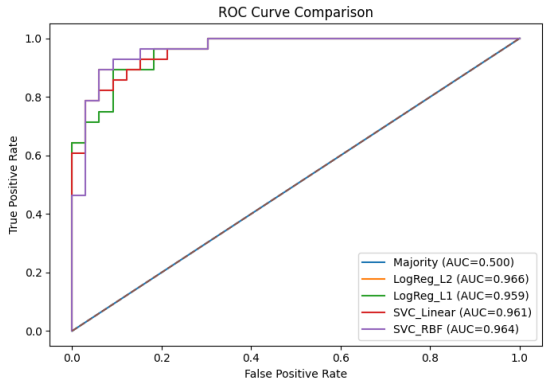

Fig. 12: ROC curves of different classification models for heart disease prediction.

Fig. 12 shows the ROC curves of the evaluated models. All machine learning models perform significantly better than the Majority classifier, whose AUC of 0.5 indicates random guessing. Among the models, Logistic Regression (L2) achieves the highest AUC (0.966), closely followed by SVC (RBF) (0.964) and SVC (Linear) (0.961). Logistic Regression (L1) shows slightly lower performance with an AUC of 0.959.

Overall, the ROC curves indicate that all models have strong ability to distinguish between patients with and without heart disease, with only minor differences in performance.

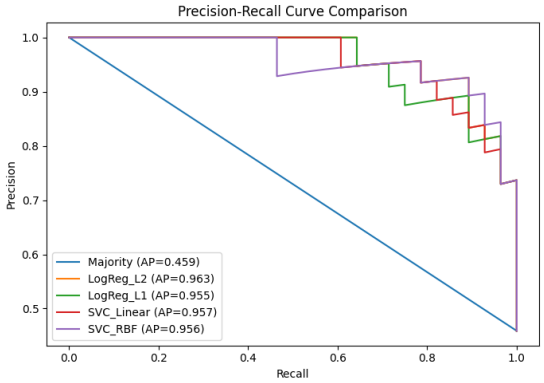

Fig. 13: Precision–Recall curves of different classification models.

Fig. 13 further supports the previous results, showing that all models outperform the Majority classifier. Logistic Regression (L2) achieves the highest AP (Average Precision),

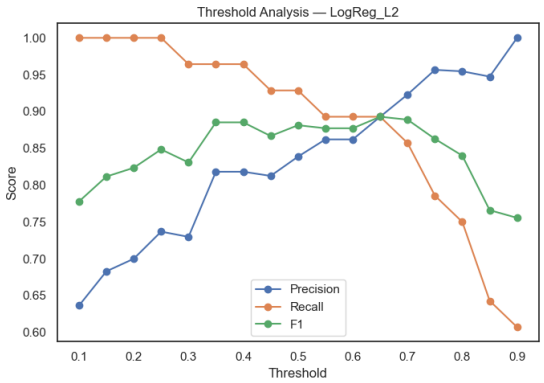

Fig. 14: Threshold Trade-off

Figure 14 evaluates different thresholds for the Logistic Regression (L2) model. As the classification threshold increases, Precision steadily improves, while Recall decreases significantly. This reflects the trade-off between correctly identifying positive cases and avoiding false positives.

The F1-score reaches its peak at around a threshold of 0.65, indicating an optimal balance between Precision and Recall. These results further reinforce the earlier observation that the model can be adjusted depending on application requirements, especially when higher Recall is prioritized in medical diagnosis.

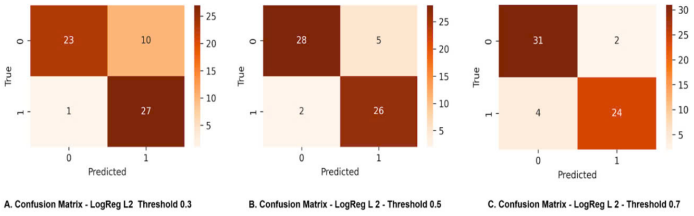

Fig. 15: Confusion Matrix at Different Thresholds

These results Fig.15 have important implications in real-world medical settings. When using a lower threshold (e.g., 0.3), the model tends to detect more patients (higher Recall), making it suitable for early screening programs. However, this also increases the number of false positives, meaning that some healthy individuals may be incorrectly identified as having the disease and require additional tests.

In contrast, when using a higher threshold (e.g., 0.7), the model reduces the number of false positives, helping to limit unnecessary tests and reduce costs. However, this also increases the risk of missing true patients, which may affect timely treatment.

Therefore, the choice of threshold should depend on the intended real-world application, balancing disease detection with cost and diagnostic risk.

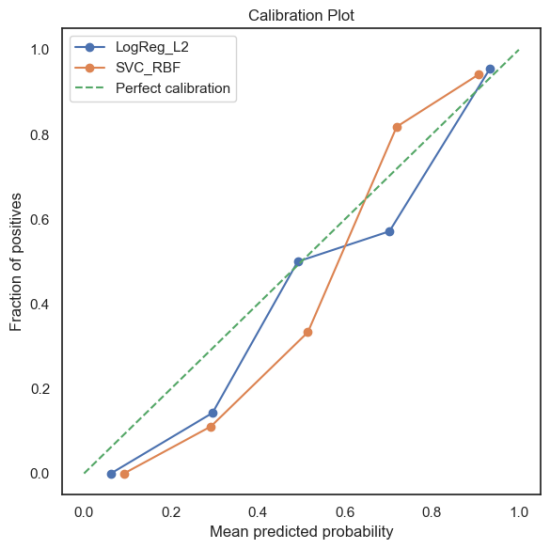

Fig. 16: Calibration Plot

Figure 16 shows the calibration curves of the models. Logistic Regression (L2) is closer to the ideal calibration line, indicating more reliable probability estimates, while SVC (RBF) shows larger deviations.

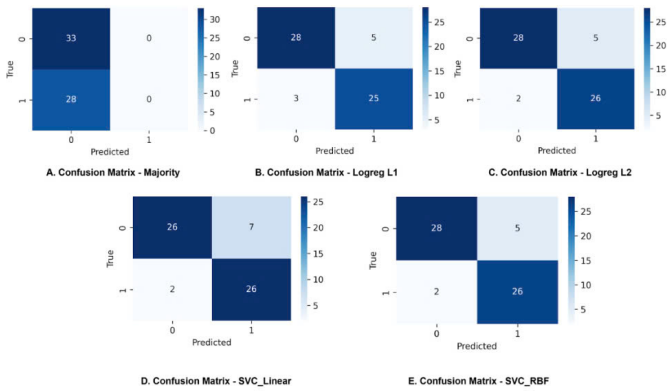

Fig. 17: Confusion matrices of the evaluated classification models.

Figure 17 shows the confusion matrices of all models. The Majority classifier predicts only the negative class, failing to detect any positive cases.

In contrast, all machine learning models significantly improve performance by correctly identifying both positive and negative samples. Logistic Regression (L2) and SVC models achieve the best balance, with high true positives and low false predictions. Overall, Logistic Regression (L2) shows the most reliable performance.

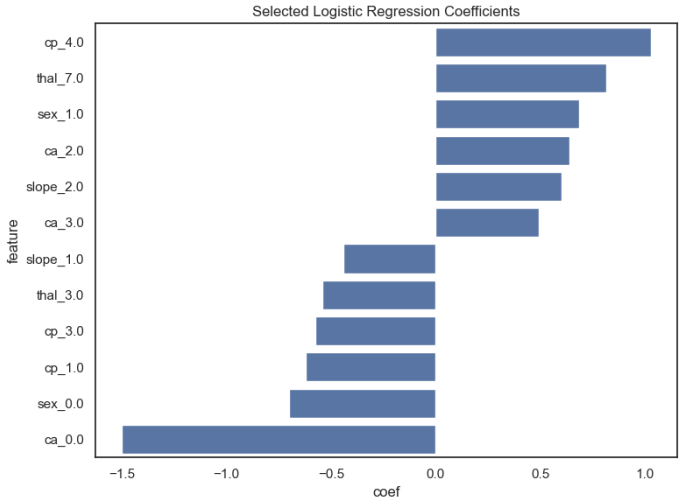

Fig. 18: Logistic Regression (L2) coefficients showing the impact of selected features on heart disease prediction.

Figure 18 shows the factors that increase or decrease the risk of heart disease. Features with positive coefficients (on the right), such as cp_4 (asymptomatic), abnormal thal, and male gender, are associated with a higher risk. In contrast, features with negative coefficients (on the left), such as cp_1 (typical angina) and female gender, are associated with a lower risk.

V. Discussion

The experimental results show that the machine learning models perform significantly better than the baseline Majority classifier. The Majority model only predicts the most frequent class in the dataset and does not utilize any feature information. As a result, it achieves an AUC score of 0.500, and the values of Precision, Recall, and F1-score are all zero. This indicates that the model cannot effectively detect patients with heart disease.

Among the evaluated models, Logistic Regression with L2 regularization achieves the best overall performance. The model reaches an Accuracy of 0.885, an F1-score of 0.881, and the highest AUC value of 0.966. This result demonstrates that the model can clearly distinguish between patients with and without heart disease. It also suggests that the relationship between the clinical features and the target variable can be effectively captured using a linear approach.

The Support Vector Classifier (SVC) models also achieve strong results. The SVC with RBF kernel obtains an AUC score of 0.964, while the linear SVC reaches an AUC of 0.961. These findings indicate that both linear and nonlinear decision boundaries are effective for this classification task.

Another important observation is the high recall value of 0.929 achieved by the best-performing models. This means that most heart disease cases are correctly detected. In medical prediction tasks, high recall is particularly important because failing to identify a positive case may lead to serious consequences.

However, the dataset used in this study contains only 303 samples, which may limit the generalization ability of the models. In addition, only traditional machine learning models are evaluated, and more advanced approaches could be explored in future work. Evaluating the models on additional datasets could also help verify the robustness and reliability of the results.

VI. Conclusion

This study applies machine learning models to predict heart disease using the Cleveland Heart Disease dataset. The problem is formulated as a binary classification task, where the models predict whether a patient has heart disease based on clinical features.

A machine learning pipeline is implemented, including data preprocessing, train–test splitting, model training, and performance evaluation. Several models are evaluated, including a Majority baseline, Logistic Regression with L1 and L2 regularization, and Support Vector Classifier with linear and RBF kernels.

The experimental results demonstrate that machine learning models significantly outperform the baseline method. Among the evaluated models, Logistic Regression with L2 regularization achieves the best performance, obtaining an AUC score of 0.966, an Accuracy of 0.885, and an F1-score of 0.881. These results indicate that the model is highly effective for heart disease prediction.

Overall, the findings confirm that machine learning approaches can provide reliable support for medical decision-making. Future work may explore more advanced models, larger datasets, and additional feature engineering techniques to further improve predictive performance and generalization capability.